

Part A · Multiple Choice

2 points each

1. **a) the percentage of water vapor the air holds relative to the maximum it can hold at that temperature**

p. 96 — because warm air can hold more water vapor than cool air, simply raising or lowering the temperature changes the relative humidity even if no water is added or removed.

2. **b) dew point**

p. 96 — the name comes from dew, which is exactly this process happening at ground level. The cold glass of ice water is the book's small-scale example.

3. **c) Cirrus**

pp. 97–98 — the prefix cirro- marks high clouds, alto- marks middle clouds, and stratus and nimbostratus are low clouds made nearly entirely of water droplets.

4. **d) moist and warm**

p. 101 — “maritime” means it formed over water and is therefore moist; “tropical” means it formed at low latitudes and is therefore warm. Continental polar (cP) is the opposite: dry and cold.

5. **a) a line of severe thunderstorms that forms along a cold front**

p. 102 — in the United States these form in spring and early summer in the Midwest, where maritime tropical air from the Gulf of Mexico meets continental polar air from Canada.

6. **b) advection fog**

p. 99 — advection fog forms from the surface contact of horizontal winds, so unlike radiation fog it can occur in windy conditions. Sea breezes then carry it onshore.

Part B · True or False

1 point each

7. **TRUE**

p. 104 — a front may become stationary when an air mass is stopped by a barrier such as a mountain range. It can bring days of rain, drizzle, and fog before breaking apart.

8. **FALSE**

With both cold and warm fronts, the air mass at the leading edge gives the front its name. A warm front marks the leading edge of moving warm air. (p. 101)

Part C · Short Answer

3 points each

9. **Denser cold air pushes underneath the warm air, forcing it upward and lowering the air pressure; this builds cumulus clouds and a line of showers or thunderstorms, with drier, clearing weather behind the front.**

Rubric: 1 pt mechanism — the denser cold air pushes up the warm air and air pressure decreases. 1 pt clouds and precipitation at the front — cumulus clouds, with winds blowing ice crystals off their tops into cirrostratus and cirrus; a line of rain showers, snow showers, or thunderstorms with blustery winds. 1 pt behind the front — the drier cold air mass arrives, precipitation stops, and skies turn clear or partly cloudy. Credit seasonal detail as a bonus: thunderstorms or tornadoes in spring and summer, heavy rain in autumn, frigid temperatures and heavy snow in winter. (pp. 101–102)

10. **An occluded front forms when a cold front catches up to a warm front, usually around a low-pressure system. From front to back the air masses are cold, warm, then cold again.**

Rubric: 1.5 pts for the mechanism — a cold front overtakes a warm front, typically around a low-pressure system, and the Coriolis effect curves the boundary toward the pole. 1.5 pts for the order: cold, warm, cold. Bonus, not required: if the third air mass is colder than both others it slides beneath them (cold occlusion); if warmer, it rides over them (warm occlusion). Weather is especially fierce right at the occlusion. (p. 103)